Ricky Faure

Principal Software Engineer

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SUMMARY

I build enterprise-grade platform infrastructure with Kubernetes Operators, Go, and Terraform. Principal Software Engineer at Optum, leading the roadmaps for Kafka-as-a-Service and Elasticsearch-as-a-Service platforms that move 20+ petabytes of data across 800+ clusters at five nines reliability, and bringing AI agents into how we build and operate them.

SKILLS

Core: Go | Python | Java | Kubernetes | Kubernetes Operators | Kubebuilder / Operator SDK | Terraform | GCP | GKE | AWS | Apache Kafka | Elasticsearch | Cassandra | Helm | CI/CD | distributed systems | SRE | infrastructure as code | technical roadmap planning

Additional: ArgoCD | Kargo | Docker | Prometheus | Thanos | Grafana | PagerDuty | GitHub Actions | canary and blue/green releases | bare-metal Kubernetes | Confluent Schema Registry | C | C# | AI agent development | responsible AI practices | Claude Code | mentorship and technical teaching

EXPERIENCE

Optum, UnitedHealth Group

JAN 2020 — PRESENT

Principal Software Engineer, TLCP — Technology Development Program Associate (Jan 2020) → Software Engineer (Jan 2021) → Senior Software Engineer (Mar 2022) → Lead Software Engineer (Feb 2023) → Principal Software Engineer, TLCP (Feb 2024 - Present)

Empower hundreds of teams across the enterprise to build data-driven services, moving tens of petabytes of data with minimal architectural overhead, using Kubernetes Operators and CI/CD to automate deployment, management, and hosting of enterprise-grade distributed systems across on-premise and multi-cloud environments.

- Platform and team leadership: Lead 15 engineers across the Kafka-as-a-Service and Elasticsearch-as-a-Service platforms and advise 2 engineering leaders , sustaining five nines reliability with zero customer data loss in production.
- Multi-year Kafka roadmap: Led the 2-year roadmap that took Kafka provisioning from CI/CD pipelines to self-service in the enterprise marketplace, backed by a custom Terraform Provider; resource deployments rose 52% and usage has grown nearly 3x .
- Executive influence: Wrote a Kubernetes security report for a VP that set the organization's GKE security standard (IAM, Workload Identity, least-privilege RBAC) and secured funding to move off bare metal; partnered with a VP to design an org-wide SRE model adopted by 4 engineering teams .
- Automated deployment pipelines: Lead the automated deployment initiative across 30 GKE clusters with ArgoCD, Kargo, promotion pipelines, and canary and blue/green releases, cutting pull requests per change from 30+ to 2 .
- Control plane architecture: Design and manage the platform's Kubernetes control plane across 800+ clusters and ~2,000 GCP instances , with the largest GKE cluster exceeding 1,000 nodes and 20+ petabytes of data movement .
- Cloud cost and GCP partnership: Delivered $2.5M in annual savings by eliminating excessive log retention in GCP; manage the GCP partnership for CUD allocation, reservations, and right-sizing.
- Agentic development: Build agentic workspaces and agent skills used by 30+ engineers , shipping ~50 fully agent-developed features to production and cutting support response time from several days to about 1 hour ; building AI support agents projected to save 4,160 engineering hours a year .
- Featured Project: Warpstream Cluster Provisioning Platform (Q4 2025 - Present)

- Architected the end-to-end design for Warpstream-based Kafka cluster provisioning and delivered a net-new Warpstream operator (Go) plus its Terraform cloud infrastructure; lead the Warpstream vendor partnership and the roadmap for it as a net-new product.
- Projected to cut beta customers' annual Kafka spend by approximately 80% ; migrated workloads already save roughly $2.9M per year .

Technologies: Go, Kubebuilder, Kubernetes Operators, Helm, Terraform, GCP, GKE, ArgoCD, Kargo, GitHub Actions, Python, Kafka, Warpstream, Elasticsearch, Prometheus, Thanos, Docker

Optum

JUN — AUG, 2017 — 2019

TDP Software Development Intern

- Returned for three consecutive summer internships, contributing business-critical features to a monolithic application with automated unit and integration testing.

Technologies: Java, Spring Boot, Groovy Spock

PROJECTS

Yo-Yo Mount Visualizer (active development)

TypeScript · React · react-three-fiber · graph modeling · pathfinding · physics simulation

3D trick engine that models yo-yo string mounts as graph topologies and aims to discover new tricks through pathfinding. Encodes mounts as schema-validated string traversals with canonical hashing, rendered in an interactive 3D visualizer with a Verlet rope physics simulation.

Spin Ledger (closed source)

Python · FastAPI · SQLAlchemy 2.0 · Alembic · rule-based entity extraction · data normalization

Single pane of glass for buy/sell/trade activity across skill toy community forums. Ingests unstructured, schemaless forum listings, normalizes them into a taxonomy-aware data model via deterministic rule-based entity extraction, and serves them through an indexed, searchable marketplace dashboard.

Self-Hosted Home Lab

Docker · Caddy · reverse proxy · TLS · DNS · VPN tunneling · NAS · Linux

Multi-machine home lab orchestrating containerized services with Docker, fronted by Caddy as a reverse proxy with automatic HTTPS; every service is served exclusively over TLS. Hosts a media server, network-attached storage, network-wide DNS-sinkhole ad blocking, secrets vaults, and closed-source project deployments, with private encrypted tunnels for secure remote access.

OSS Feed

JavaScript · GitHub Actions · RSS/Atom · GitHub Pages

Self-hosted weekly RSS digest that tracks releases from curated open source projects. A watchlist YAML feeds GitHub Actions to fetch releases, generate an XML feed, and publish via GitHub Pages, with Slack/Discord webhook notifications.

HostPort Allocator (open source contribution)

Kubernetes

Contributed to HostPort Allocator, an open source Kubernetes host port allocator.